

ALEXANDER IMMANUEL GOERKE ZAMORA

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SUMMARY

Industrial Electronics and Automation engineer from UPV, with an Erasmus+ year at DTU (Copenhagen). My focus is robotics, digital control, and signal processing, from theoretical derivation through simulation to real hardware implementation and experimental validation, in international and multidisciplinary environments.

EDUCATION

IALE INTERNATIONAL SCHOOL

Secondary Education (ESO & Bachillerato)

- Awarded the Science Prize for academic excellence.

UNIVERSITAT POLITÈCNICA DE VALÈNCIA (UPV)

B.Sc. Industrial Electronics and Automation Engineering

TECHNICAL UNIVERSITY OF DENMARK (DTU)

Erasmus+ Exchange

Valencia, Spain

2016 – 2022

Valencia, Spain

2022 – 2026

Lynby, Denmark

2025 – 2026

EXPERIENCE

CEBREO MEDICAL A/S

Software Engineer

Copenhagen, Denmark

Jan 2026 – Jun 2026

- Developed signal-processing algorithms and visualization tools for biomedical signals (EEG, PPG, and triaxial accelerometer data), including filtering, pulse-analysis, and posture-detection algorithms based on accelerometer signals, supporting clinical biosignal inspection across EDF, CSV, and .mat data formats.
- Designed synthetic test-signal pipelines for algorithm validation and QA, working within a multidisciplinary team of software engineers, physiologists, and product developers.

PRIVATE TUTORING

Maths, Physics & Chemistry Tutor

Valencia, Spain

2022 – Present

- Provide one-to-one tutoring in mathematics, physics, and chemistry for secondary school students since 2022, planning and delivering personalized study plans and applying structured problem-solving to improve academic outcomes.
- Recommended by school teachers for consistently strong academic results, with a focus on clear communication and mentoring.

PROJECTS

BACHELOR THESIS: DELTA ROBOT

DTU / UPV, 2025–2026

- Designed the control system for a delta robot supporting a 5-year PhD programme at DTU on agricultural weed-intervention, deriving full IK/FK kinematics, building a complete simulation suite in Python, and porting the controller to C++ deployed on a Teensy 4.1.
- Validated feedforward, smooth bounded feedback, and full PID strategies, achieving 1–2 mm end-effector positioning accuracy through quantitative real-robot experiments.

DIFFERENTIAL-DRIVE ROBOT

DTU Digital Control, 2025

- Ranked 1st in class. Designed PI/PD controllers with encoder-based sensor fusion, validated in Simulink and deployed on a Teensy 4.1.

VISION-GUIDED 4-DOF ROBOT

DTU Robotics, 2025

- Developed the vision-to-motion pipeline for a 4-DOF robotic arm, converting pixel coordinates to world space and reaching targets via inverse kinematics, with HSV segmentation, camera calibration, and Dynamixel servo control.

TECHNICAL SKILLS

- Engineering:** Robotics, Digital Control, Signal Processing, Embedded Systems, Circuit Design, IK/FK Kinematics, Sensor Fusion, Mechatronics
- Programming:** Python, C/C++, MATLAB, Arduino, HTML/CSS
- Design & Simulation:** SolidWorks, AutoCAD, MATLAB/Simulink, Proteus, OrCAD
- Tools & AI:** Git, VSCode, LaTeX, Claude, Codex

CERTIFICATIONS & AWARDS

- ISO 21500 & ISO 21502: Project, Programme and Portfolio Management (Danish Standards, Jan 2026)
 - Advanced English (C1) | German (A2) | Web Programming (HTML/CSS/JS) | Energy Systems (CFP, UPV, 2024)
- Languages:** Spanish (native) | English (C1) | German (A2)